Mastering TMDL in Power BI

The Code-First Revolution

```
model {
  compatibilityLevel: 1601
  culture: "en-US"
}
```

From clicking to scripting.

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SPEAKER

Davide Lambertini

Business Intelligence Engineer @ Lucient Italia

Lucient

Professional Background

- **10+ years** working with data from different perspectives
- Started as **Business Analyst** in the **Retail Industry**
- Transitioned into **Data Consultancy**

Interests

- Addicted to **data & tech**
- Passionate about **economy** and **finance** topics

Community

- Occasional **speaker** at data community conferences

Personal

- Love **traveling abroad**!

Microsoft CERTIFIED POWER BI DATA ANALYST ASSOCIATE

Microsoft CERTIFIED FABRIC ANALYTICS ENGINEER ASSOCIATE ASSOCIATE

Contact

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The challenges of Traditional GUI-based PowerBI development

Manual and Repetitive Changes

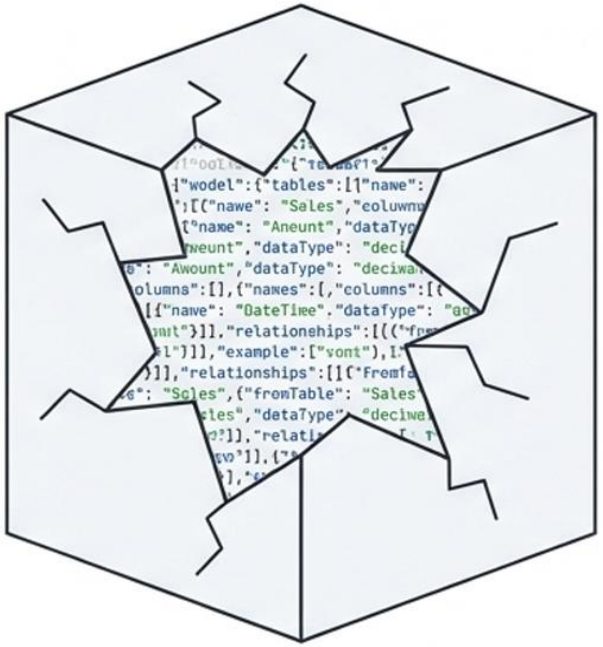
No Version Control

Difficulty in Reuse

Low Model Transparency

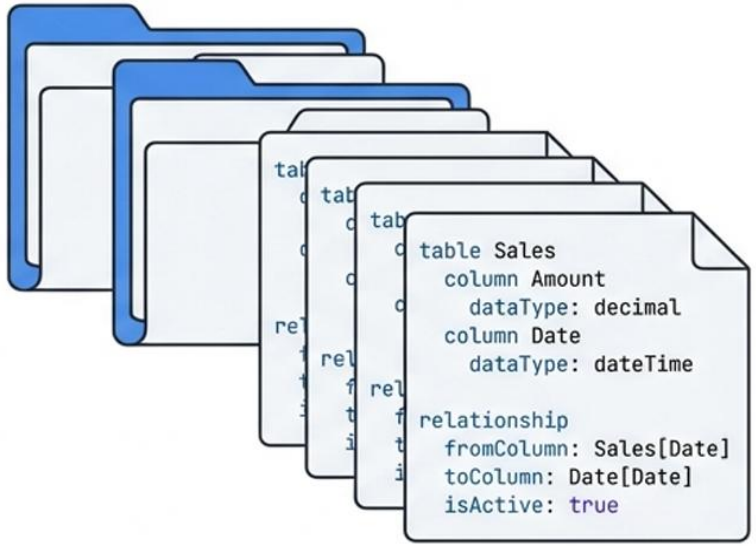
TMDL: Tabular Model Definition Language

TMSL (JSON) - Monolithic & Complex



- Monolithic & Fragile
- Machine-Readable Only
- Merge Conflicts

TMDL - Modular & Human-Readable



- Human-Readable (YAML-like)
- Modular File System
- Git-Ready

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Why should I care about TMDL ? Isn't it only Versioning?

Speed

Massive changes, regex and copy-paste reduce development-times by up to 50%.

Control

Granular access to hidden properties from the UI (Storage Mode, Detail Rows, Summarization).

Collaboration

Structure with `multiple files` and `diffs` enable true Git workflows without conflicts.

Quality

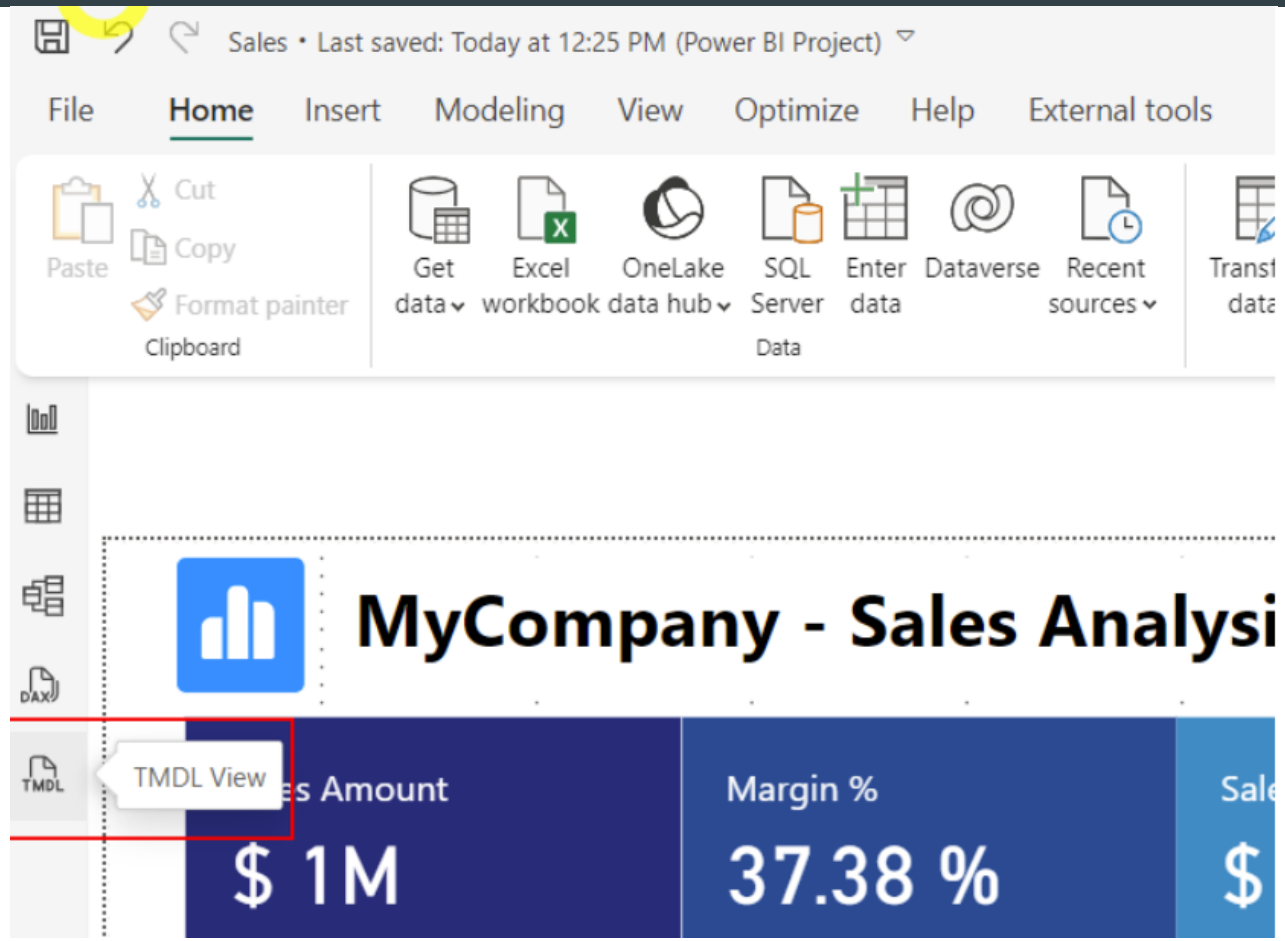
Standardization through `reusable scripts` and automated documentation via `AI`.

Get Started with TMDL View

TMDL View is a feature in PowerBI Desktop, Released in Jan 2025 and promoted GA in Autumn 25 .

It allows you to render your TMDL code, script and perform metadata editing on semantic model directly via code instead of using UI.

If you have old versions of Powerbi, you may have to switch on the preview feature of TMDL View from Options->Preview Features

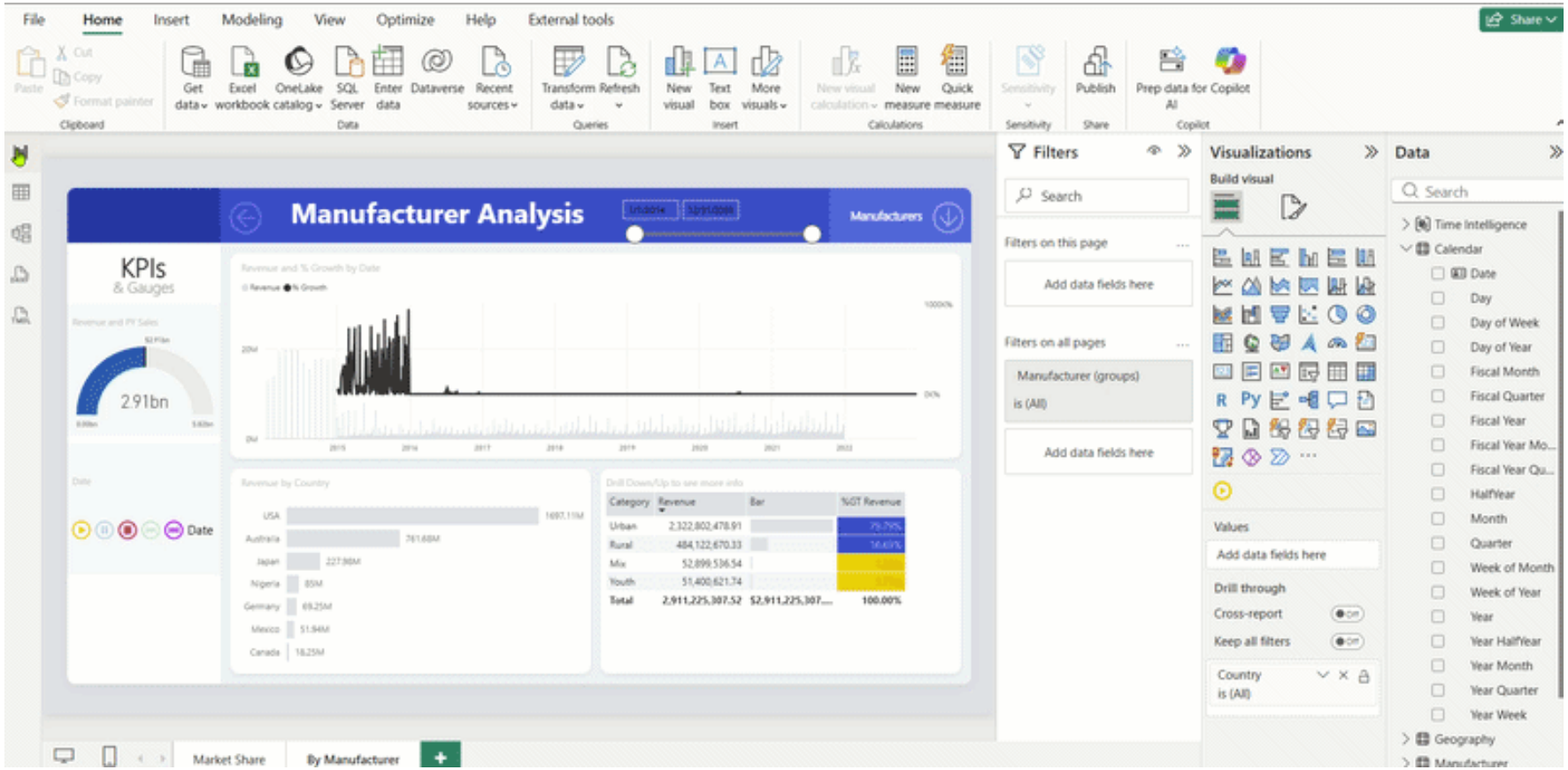


Get Started with TMDL View

Exploring Semantic Model Metadata

Even if you're not looking to modify your semantic model, TMDL View is a powerful tool for exploring and understanding model metadata.

You can explore any particular measure, column, table or complete Semantic Model. You can either drag and drop the field to the space or simply right click on it and select script tab



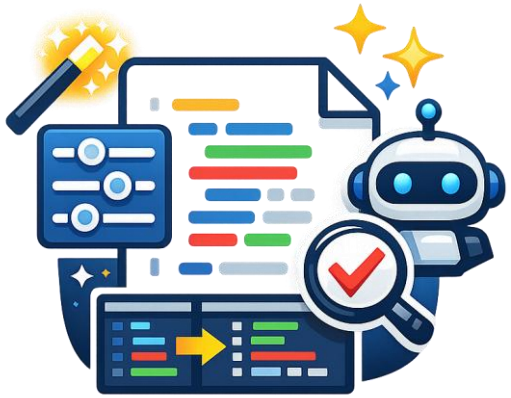
Features of TMDL VIEW

TMDL View's code editor isn't just for scripting—it's a **comprehensive development toolkit** easy to use and just inside PowerBI. Key features include:

- Visual Clarity: **Semantic Highlighting** and **Formatting (Shift+Alt+F)** distinguish keywords, names, and properties for immediate readability.
- **Safety First:** The **Preview Pane** provides a side-by-side "diff" view (Red/Green) to validate changes before clicking **Apply**.
- **Coding Power:** **Intellisense** and **Quick Fixes** (Lightbulb icon) provide smart suggestions and resolve errors like duplicate lineage tags instantly.

Multi-Cursor Editing (CTRL+SHIFT+L) mode enables Monaco-like editor experience.

- **Precision:** **Error Diagnostics** use "squiggles" and a **Problems Pane** to navigate directly to syntax issues.
- **The Golden Rule:** TMDL is strictly **indentation and hierarchy sensitive** (YAML-style).



TMDL View is not a replacement for Tabular Editor

Tabular Editor

Best for Complex Automation (C# Macros) & Best Practice Analyzer. External Tool.

Use Case: Automazione Complessa

TMDL View (Native)

Best for Daily Dev, Visual Regex, & IntelliSense. Native Integration.

Use Case: Sviluppo Quotidiano

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Key Use Cases of TMDL VIEW

TMDL PARADIGM SHIFT

TMDL (Tabular Model Definition Language) is a human-readable, folder-based representation of the Power BI semantic model, enabling bulk updates, automated documentation, and professional DevOps workflows.

HYPER-EFFICIENT MODEL MANAGEMENT

Find, Replace, and Multi-Cursor Editing

Regex

s/Measure[A-Z]/New_Measure/g

```
variable varials = {
    Measure(D-2)/New_Measure,
    Measure(A-2)/New_Measure,
    Measure(D-2)/New_Measure,
    Measure(B-2)/New_Measure,
    Measure(A-2)/N
};
```

Use Regex and multiple cursors a Multi-Cursor Editing

Bulk Property and Metadata Updates

Script-driven execution

Unassigned → Sales Measures

```
move Measures/Salec" to
    Folder/"Sales Measures"
```

Update M-Code Without the Query Editor

TMDL.ed

```
let Source = "SalesData",
    "Sato: "Model_Definitio
    "Sorece": "Saals-Iomata
    "italmoate": "sonrige-t
);
```

Low loading...

Directly Modify Power Query Scripts

PROFESSIONAL WORKFLOWS & REUSABILITY

Pattern Reuse and Instant Object Creation

Library

- Calendar Table ...
- Calculation Group ...
- Time Intelligence Measures

Instantly placed in a source file (copy pasted) into a new TMDL model project

Structural Flexibility and Connection Toggling

```
connection: import
```

Instantly toggled

```
connection: directquery
```

Import → DirectQuery

Instantly toggle if code and Connection Toggling

CI/CD and Collaborative Development

Deployment Pipeline

Build → Test → Deploy

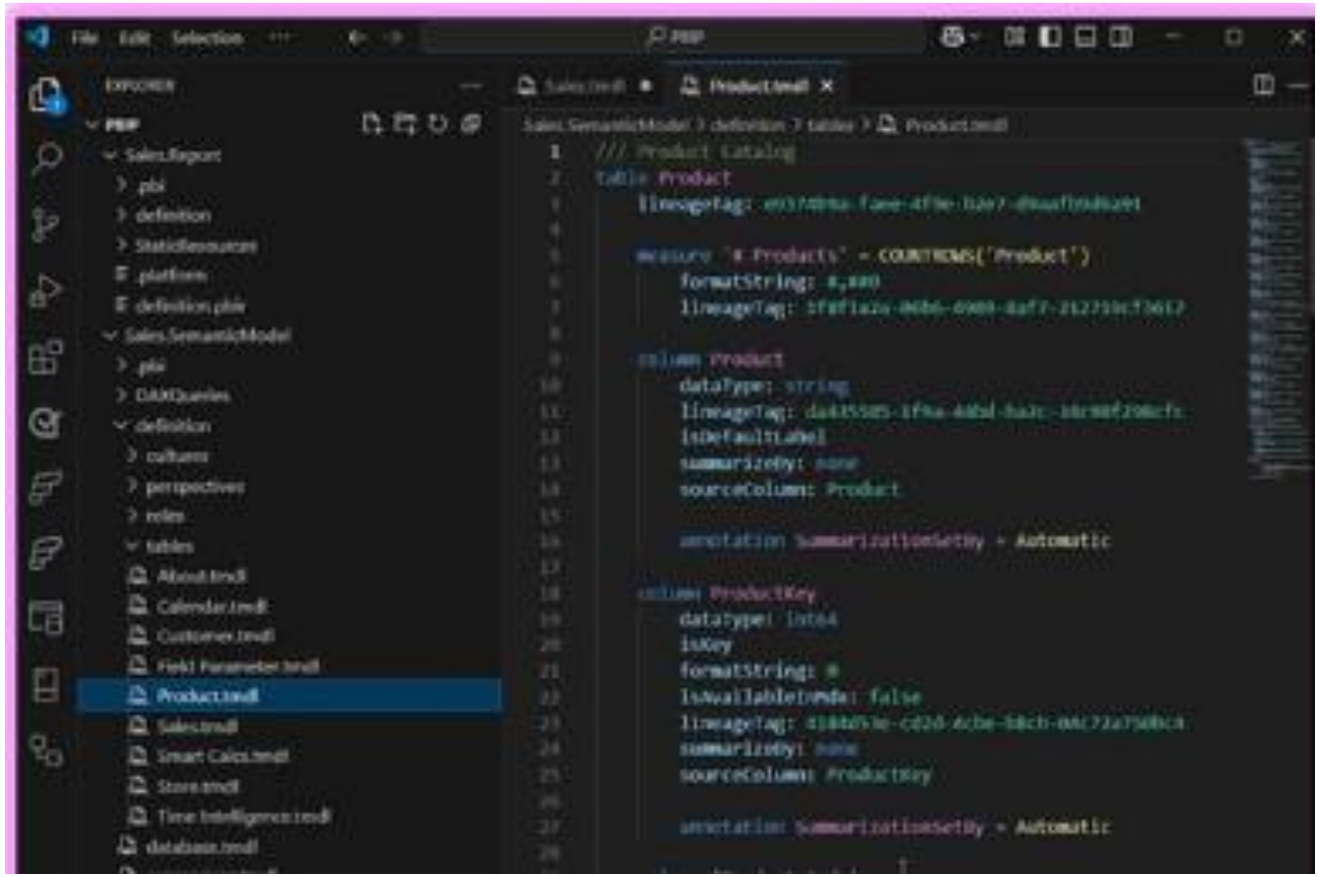
Enable Professional Version Control (Git) and Automated Deployment Pipelines

TMDL in VS Code Experience (PBI Dev Mode)

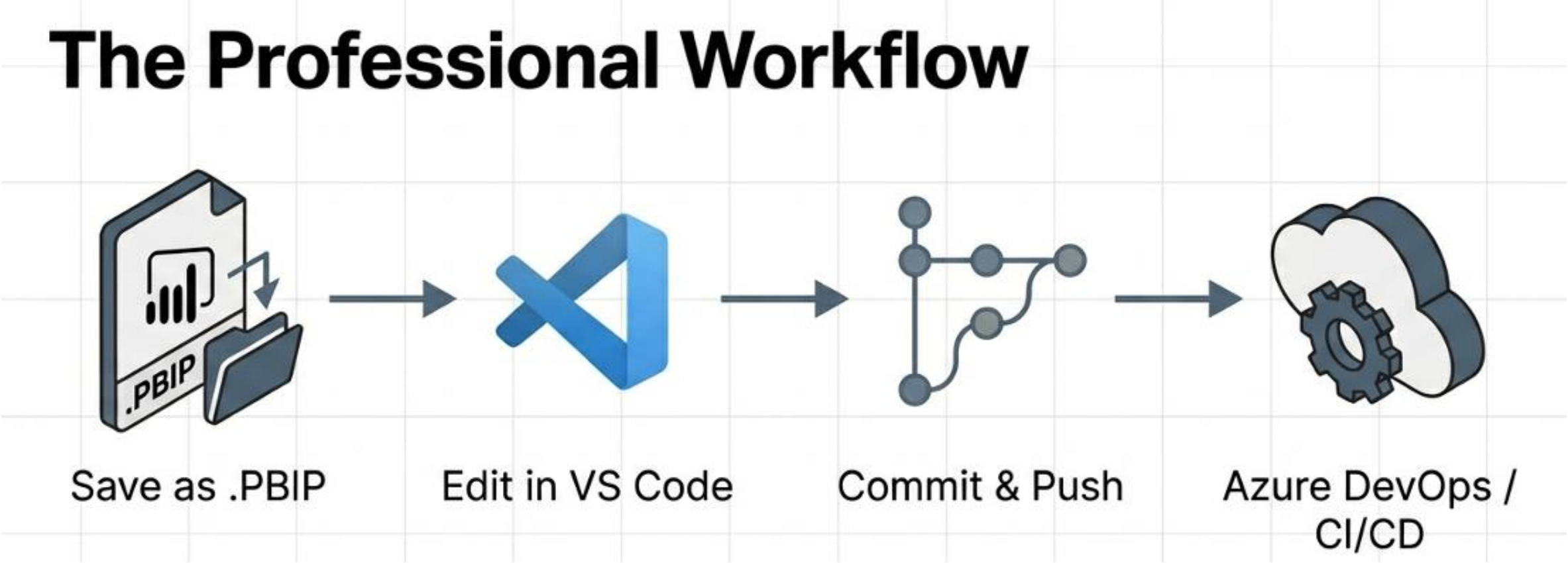
- VS Code supports TDML through the TMDL Extension
- For a professional, full code experience in Powerbi, with TMDL enabled save your pbix as a pbip (pbi project) and open it in VS Code
- you will get the entire TMDL semantic model definition + Report definition everything split into single, atomic, organized files ready for source control!

VS Code is the Pro Dev tool so you will get all the features already presented in TMDL View +

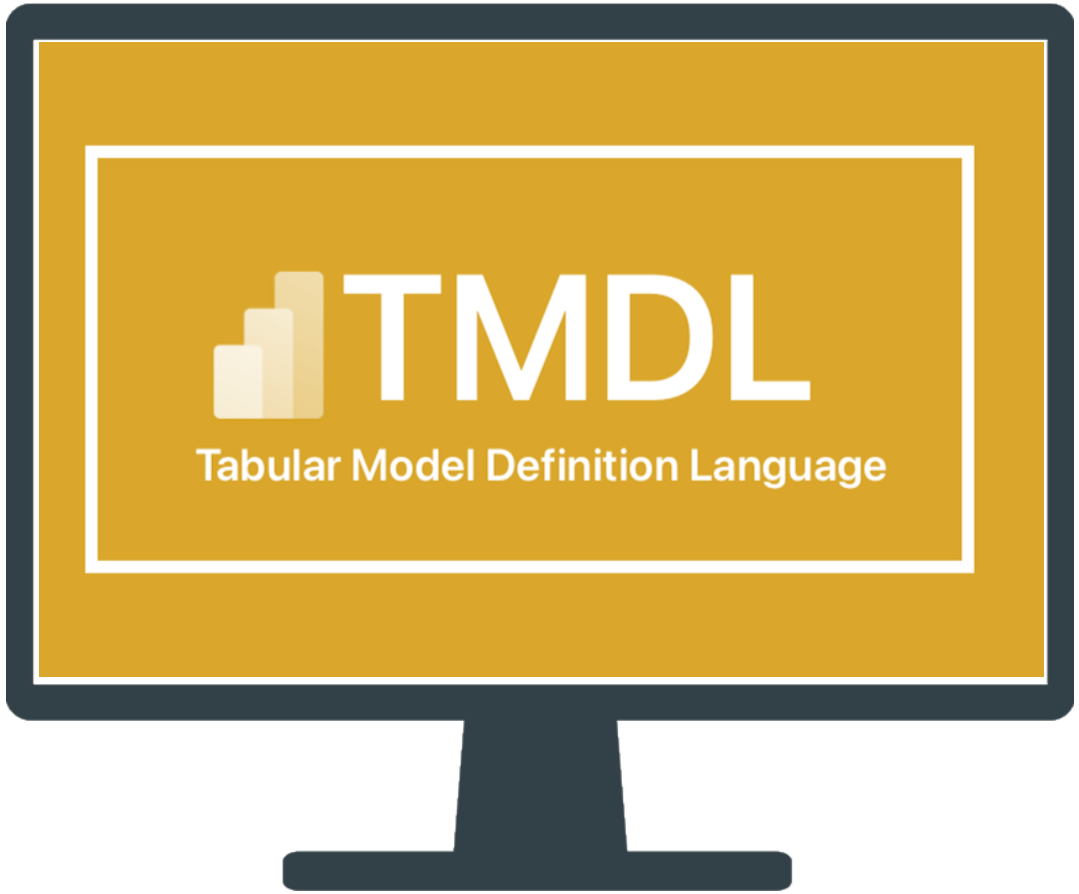
- Full Git/Devops Integration for CI/CD workflows
- AI Agent assisted development with Github Copilot
- And many more…



TMDL in VS Code Experience (PBI Dev Mode)

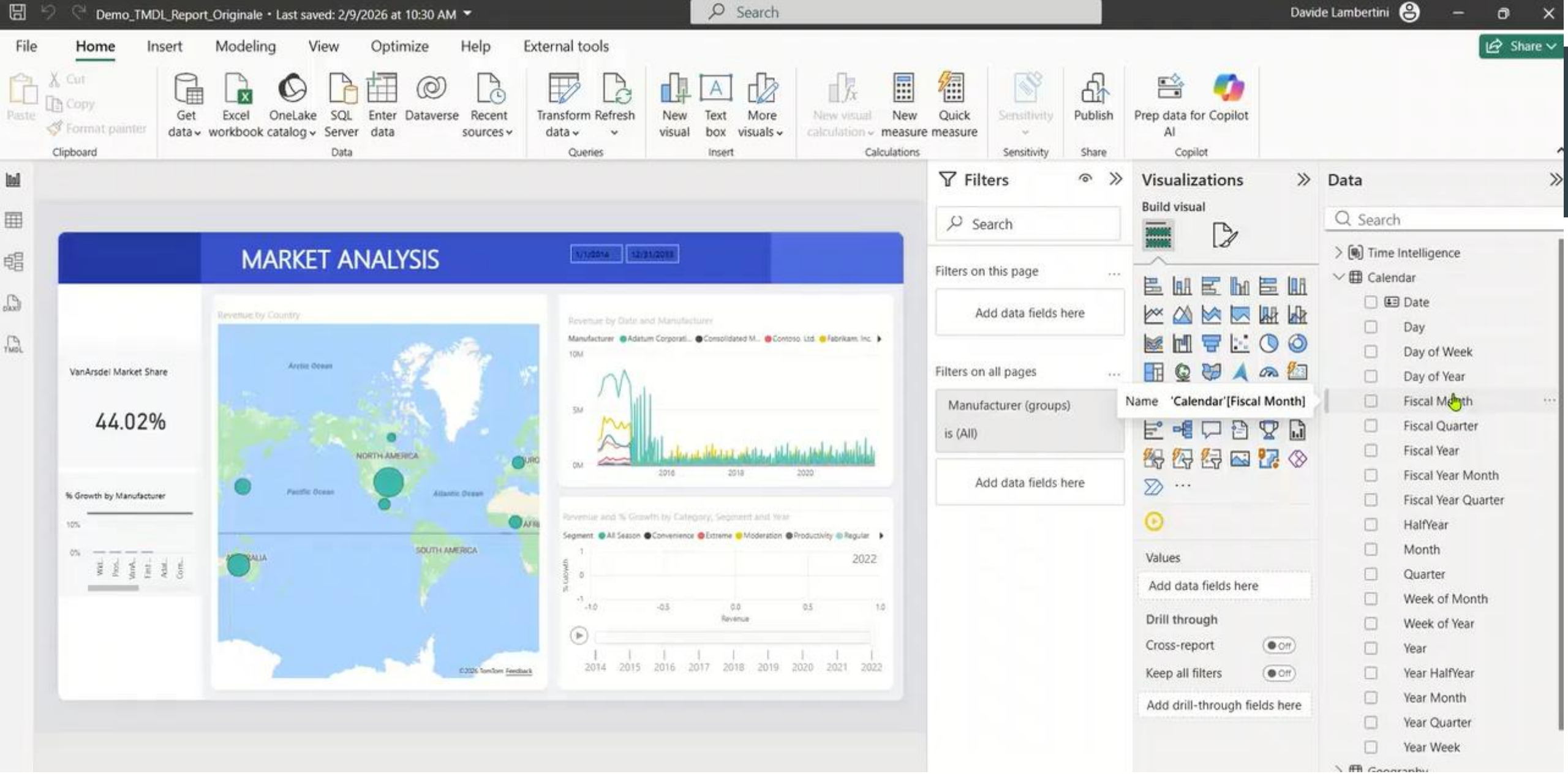


Feature Demo 1 – Regex Replace Col names



[Brand Name]
[Product Category]
[Total Cost]

[BrandName]
[ProductCategory]
[TotalCost]


Demo_TMDL_Report_Originale • Last saved: 2/9/2026 at 10:30 AM
Search
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File
Home
Insert
Modeling
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Optimize
Help
External tools
Share
Cut
Copy
Paste
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Prep data for Copilot AI
Copilot
MARKET ANALYSIS
VanArsdel Market Share
44.02%
% Growth by Manufacturer
Revenue by Country
NORTH AMERICA
SOUTH AMERICA
Revenue by Date and Manufacturer
Revenue and % Growth by Category, Segment and Year
2022
Filters
Search
Filters on this page
Add data fields here
Filters on all pages
Manufacturer (groups)
is (All)
Add data fields here
Visualizations
Build visual
Name 'Calendar'[Fiscal Month]
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Time Intelligence
Calendar
Date
Day
Day of Week
Day of Year
Fiscal Month
Fiscal Quarter
Fiscal Year
Fiscal Year Month
Fiscal Year Quarter
HalfYear
Month
Quarter
Week of Month
Week of Year
Year
Year HalfYear
Year Month
Year Quarter
Year Week

Warning: Regex Patterns in TDML can be tricky ☹

1. Lack of Structural Awareness
Regex does *not* understand TMDL's hierarchical/indented YAML-like structure.
This can lead to ambiguous or unintended matches.
Always validate results before applying.
2. Limitations of the Power BI Regex Engine
Power BI uses a simplified .NET regex engine in TMDL View.
It Lacks advanced features found in PCRE, Perl, and full IDEs.
Features like recursion and robust handling of nested TMDL elements are **not supported**.
3. Challenges Using LLMs
LLMs can help generate regex, but Regex world has many syntaxes, and dialects.
Without having a good understanding of Regex coding yourself, you can spend hours debugging wrong patterns that "should" be simple.

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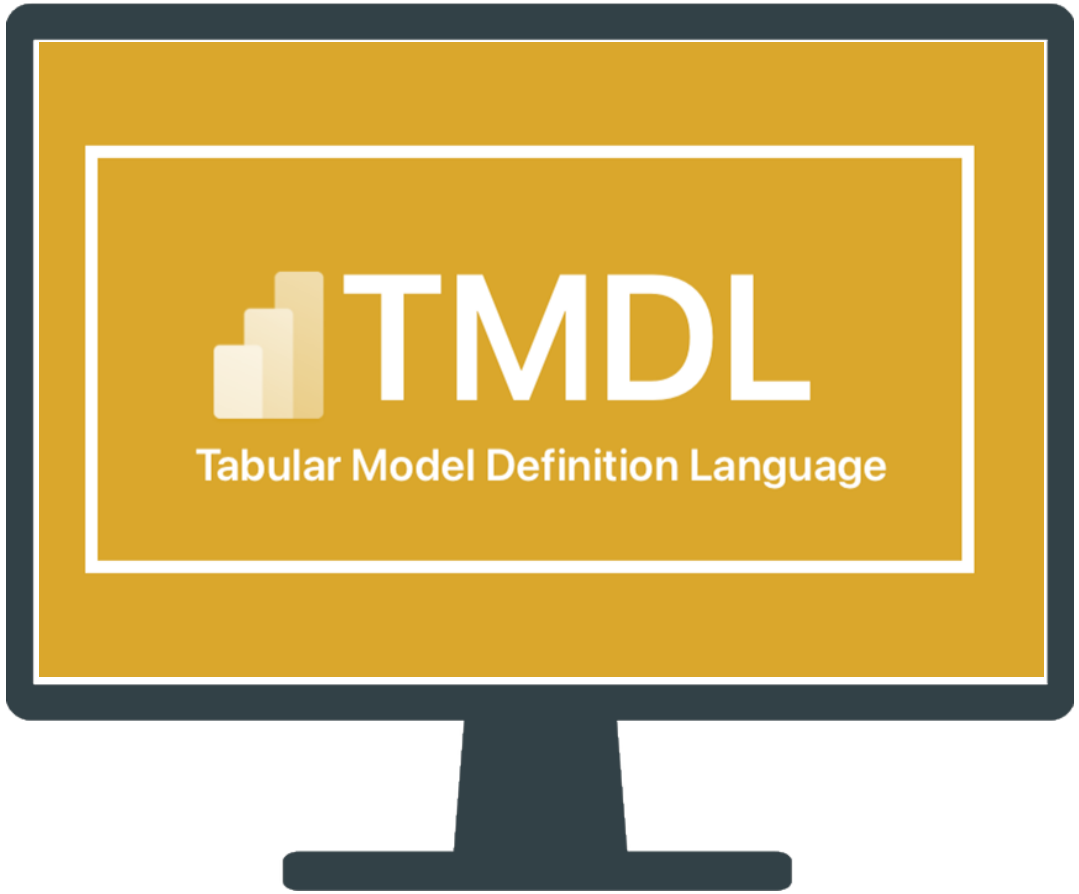
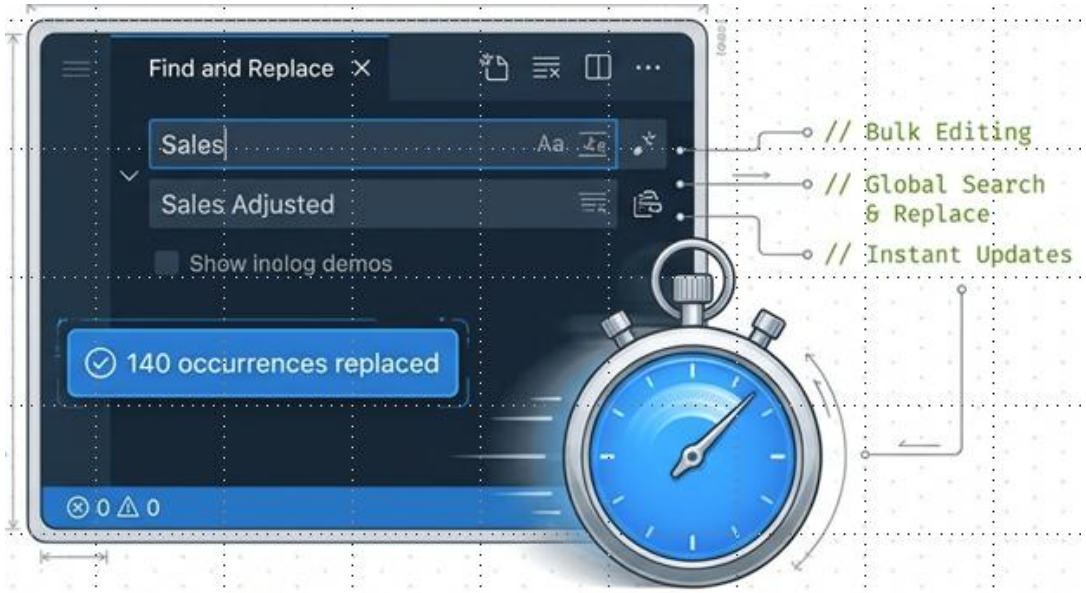
Feature Demo 2 - Move measures to Table





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Share

Paste Cut Copy Format Find Replace Command palette

Clipboard Editing

Apply Preview Share feedback

```
1   createOrReplace
3       model Model
12          table Sales
20
21          measure 'VanArsdel Market Share' = DIVIDE([VanArsdel Sales],[Sales])
22              formatString: 0.00%;-0.00%;0.00%
23              lineageTag: d488b6d9-8877-4c0c-bd9d-2bc1c1d4c9c2
24
25          measure 'PY Sales' = CALCULATE(SUM(Sales[Revenue]),SAMEPERIODLASTYEAR('Calendar'[Date]))
26              formatString: \$#,0.00;(\$#,0.00);\$#,0.00
27              lineageTag: c0640586-f8d1-4f76-8eec-325fd4b637fc
28
29              annotation PBI_FormatHint = {"currencyCulture":"en-US"}
30
31          measure '% Growth' = DIVIDE(SUM(Sales[Revenue])-[PY Sales],[PY Sales])
32              formatString: 0.00%;-0.00%;0.00%
33              lineageTag: f2c83fef-957a-4c23-af3d-53aa0c0ab8b3
34
35          measure Bar = SUM(Sales[Revenue])
36              formatString: \$#,0.###############;(\$#,0.###############);\$#,0.###############
37              lineageTag: f06ca870-193e-4fda-85a4-feb62955989b
38
39              annotation PBI_FormatHint = {"currencyCulture":"en-US"}
40
41          measure Sales = SUM(Sales[Revenue])
42              formatString: \$#,0.###############;(\$#,0.###############);\$#,0.###############
43              lineageTag: 26364afe-add6-42ee-92f2-3400a17f5a6e
44
45              annotation PBI_FormatHint = {"currencyCulture":"en-US"}
```

Problems 0 Output

Backup Create Measure Table

Data

Tables Model

View and organize all of the items in your semantic model. Learn how

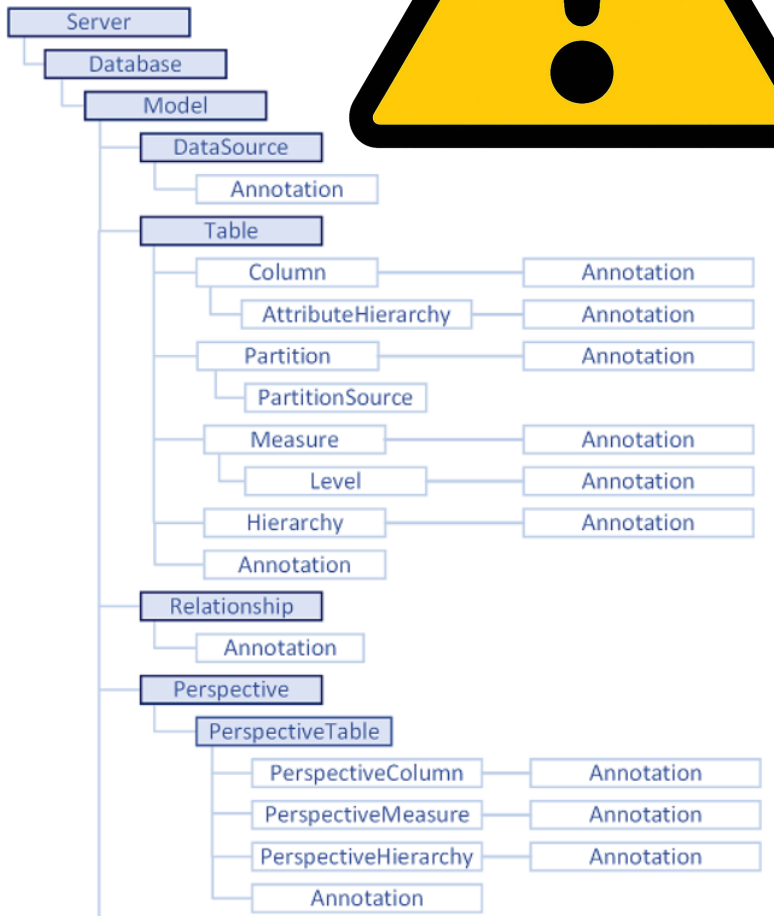
Search

- Geography
- Manufacturer
- Product
- Sales
  - Calendars (0)
  - Columns (5)
    - CountryZip
    - Date
    - ProductID
    - Revenue
    - Units
  - Hierarchies (0)
  - Measures (6)
    - % Growth
    - Bar
    - PY Sales
    - Sales

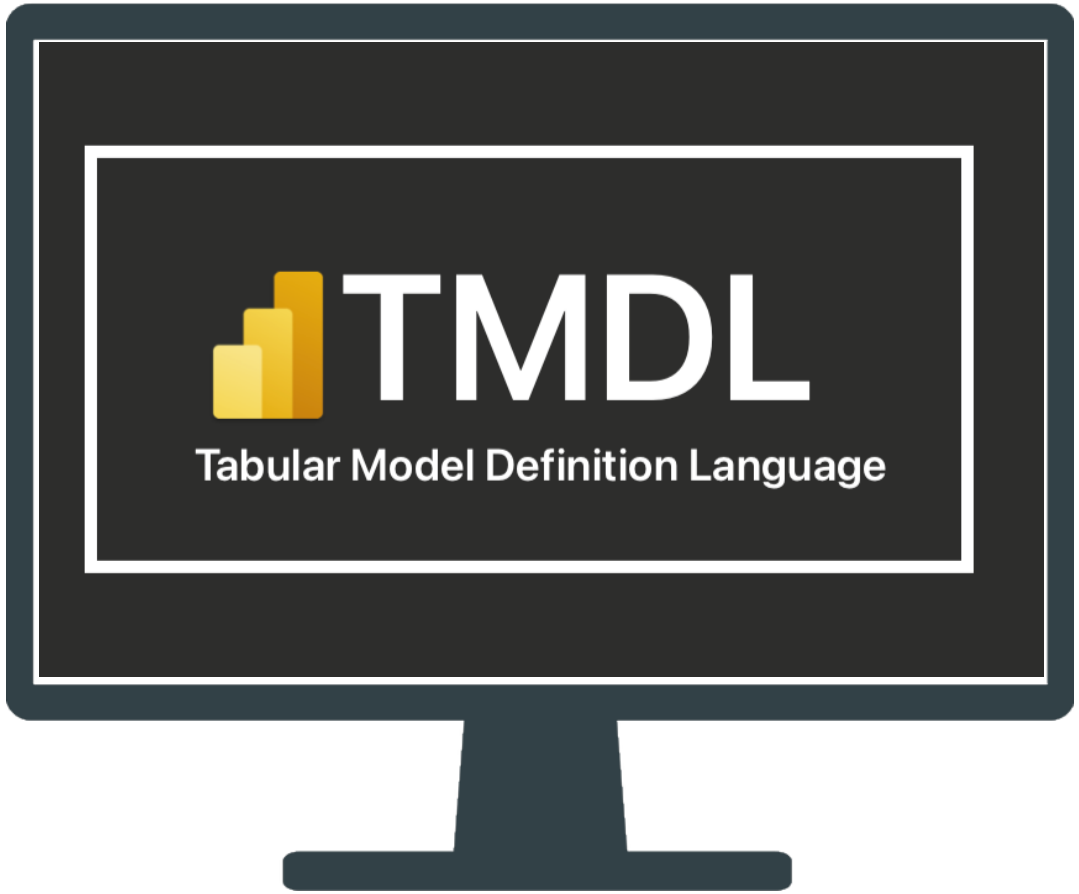
Warning: Moving objects changes IDs

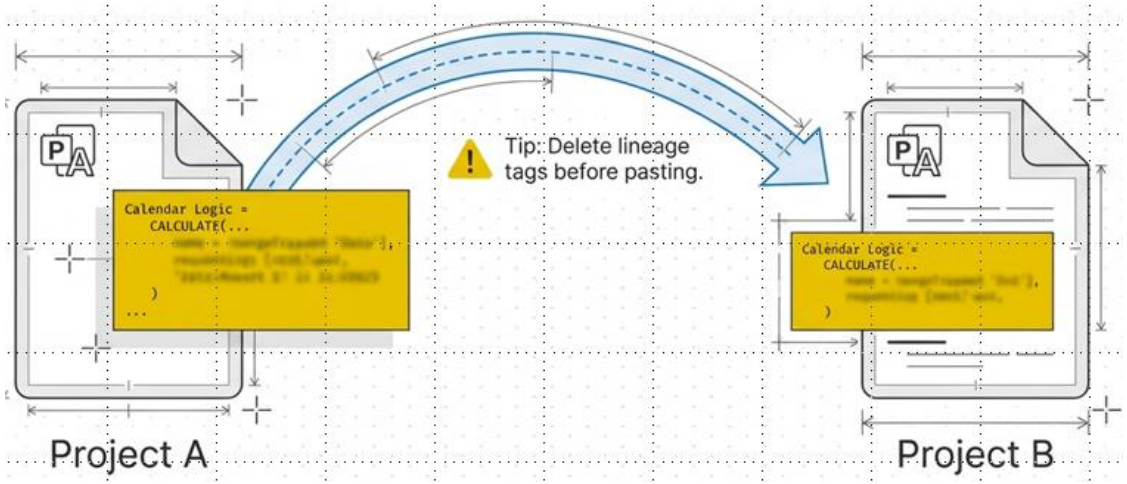
- When moving or renaming objects, dependent objects require scripting the entire parent.
- TMDL View always commits changes in a single transaction.
- You cannot create or replace objects using the same lineageTag or name within the model.
- TMDL View often performs drop & recreate operations — visible in the Preview. If you see a delete, consider the impact on your visuals
- lineageTag ≠ internal visual identity: it's for data lineage and governance, not for binding visuals.
- Power BI Desktop behaves differently under the hood and can preserve internal IDs.

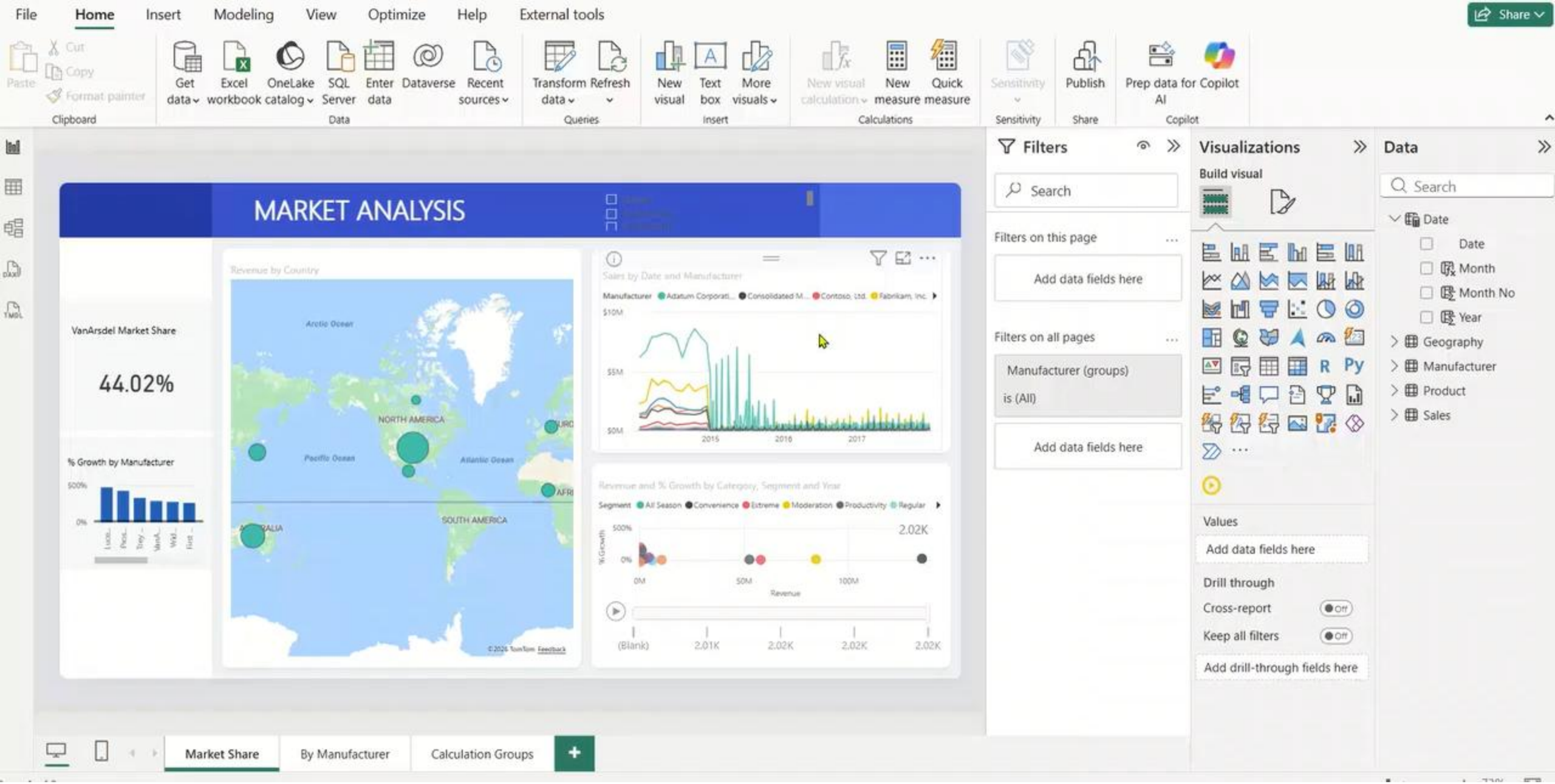
Tip: try TE3's "Move to Table" feature.



Feature Demo 3 – Reusable Scripts & Patterns






File
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External tools
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
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New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
MARKET ANALYSIS
VanArsdel Market Share
44.02%
% Growth by Manufacturer
Revenue by Country
Arctic Ocean
NORTH AMERICA
Pacific Ocean
Atlantic Ocean
SOUTH AMERICA
Sales by Date and Manufacturer
Manufacturer
Adatum Corporati...
Consolidated M...
Contoso, Ltd.
Fabrikam, Inc.
$10M
$5M
$0M
2015
2016
2017
Revenue and % Growth by Category, Segment and Year
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Extreme
Moderation
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Filters
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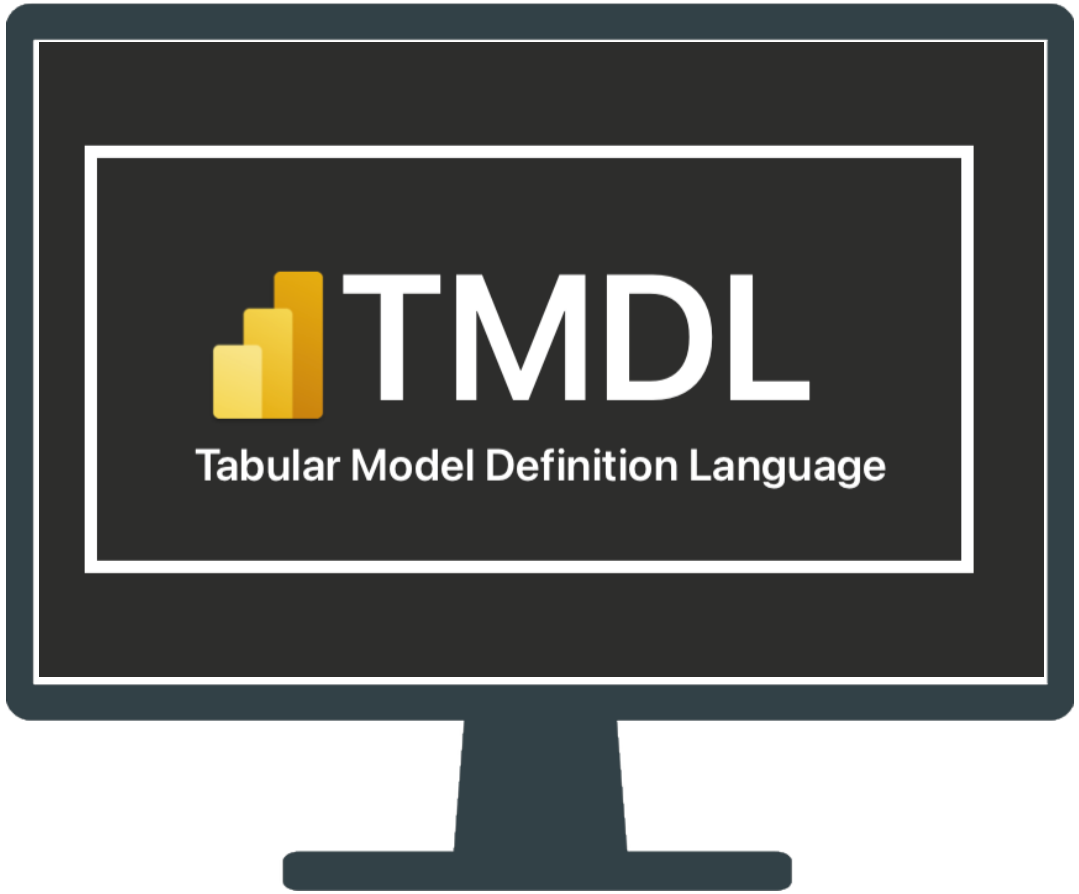
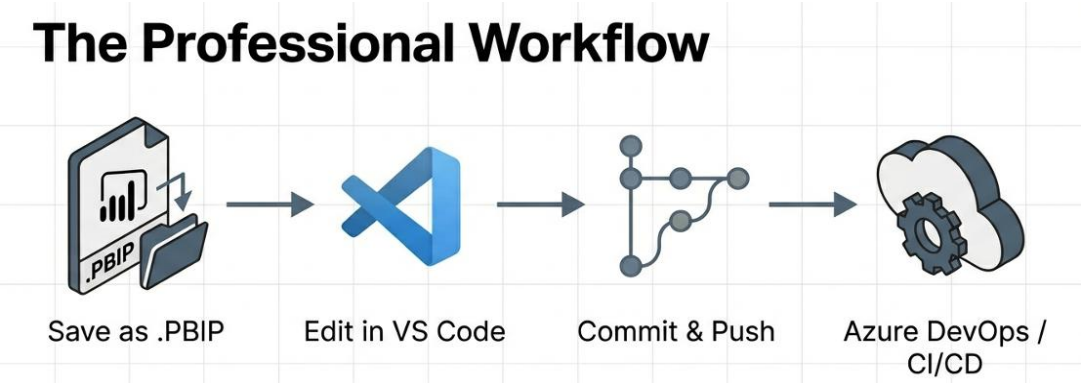
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Date
Date
Month
Month No
Year
Geography
Manufacturer
Product
Sales
Market Share
By Manufacturer
Calculation Groups

Reusability Patterns is a great use case for TMDL ☺

- Scripting tables boosts productivity and supports standardization across semantic models, improving reusability and consistency.
- TMDL preserves model-level properties (e.g., SortByColumn) that pure M code cannot capture.
- Not all model properties are directly exposed, while others—like disabling Auto Date/Time—aren't available in Power BI's model features.
- Warning: Reusing shared templates (date tables, calculation groups) can cause conflicting lineage when copying objects across models. Always remember to Remove lineageTag to safely reuse components.

Use Case 1 – CI/CD Professional Workflow





Procurement • Last saved: Today at 1:12 PM (Power BI Project)

Search

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File | Home | Insert | Modeling | View | Optimize | Help | External tools | Format | Data / Drill

Share

Paste | Cut | Copy | Format painter — Clipboard

Get data | Excel workbook | OneLake catalog | SQL Server | Enter data | Dataverse | Recent sources — Data

Transform data | Refresh — Queries

New visual | Text box | More visuals — Insert

New visual calculation | New measure | Quick measure — Calculations

Sensitivity — Sensitivity

Publish — Share

Prep data for Copilot AI — Copilot

p 4	Group 5	Group 6	CCA
amento lappatura (4000)	Trattamento lappatura (5000)	Trattamento lappatura (6000)	CODICE DISPONIBILE (M568)
abrasive (973)	Mole abrasive (1045)	Mole abrasive (1091)	MOLE ABRASIVE (M389)
(971)	Mole (1043)	Mole (1089)	MOLE (M387)
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rt (1832)	Fickert (1973)	Fickert (2095)	FICKERT (H990)
hing coatings (1833)	Polishing coatings (1974)	Polishing coatings (2096)	POLISHING COATINGS (H991)
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ring Tools (1835)	Squaring Tools (1976)	Squaring Tools (2098)	SQUARING TOOLS (H993)
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dri Rotocolor (1645)	Cilindri Rotocolor (1792)	Cilindri Rotocolor 7 (2000)	CILINDRI ASTRA 48 (H288)

Filters

Search

Filters on this visual

- CCA is (All)
- Commodity is (All)
- Group 1 is (All)
- Group 2 is (All)
- Group 3 is (All)
- Group 4 is (All)
- Group 5 is (All)

Visualizations

Build visual

Columns

- Commodity Cod Hier...
  - Commodity
  - UnderCommodity
  - Group 1
  - Group 2
  - Group 3
  - Group 4
  - Group 5
  - Group 6
  - CCA

Drill through

Data

Search

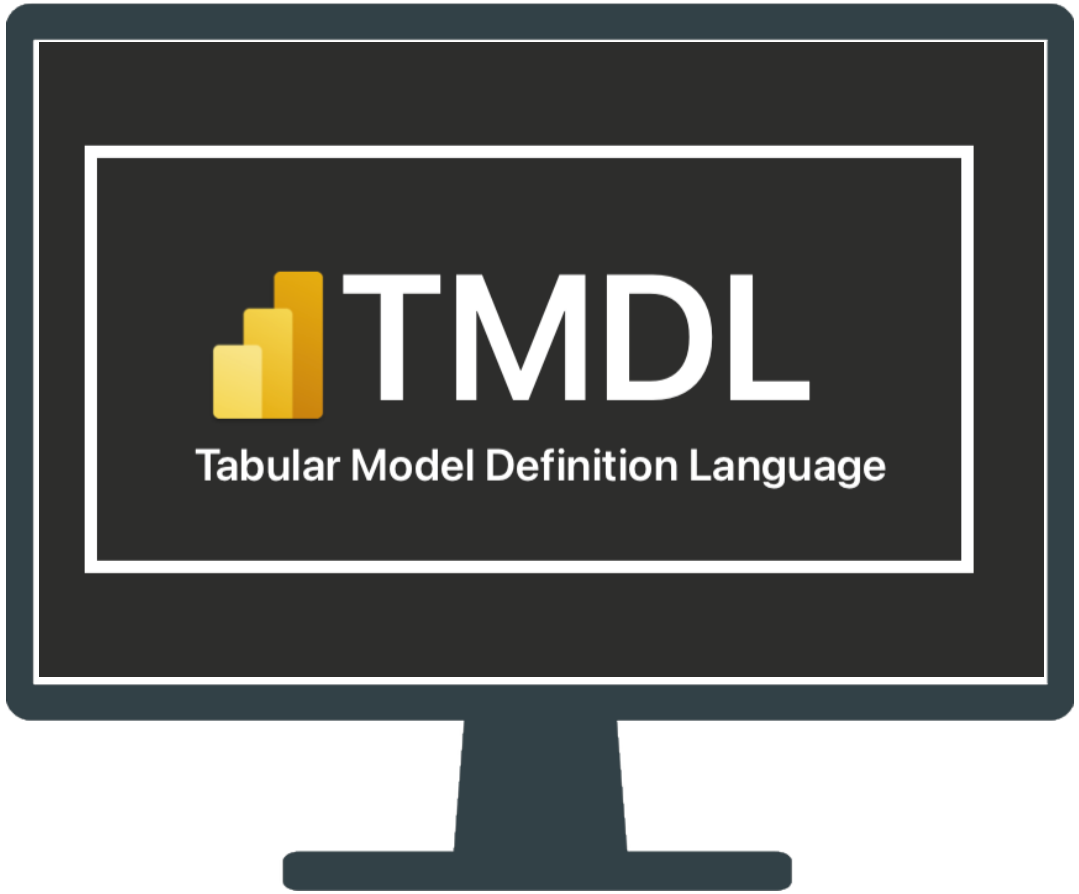
- UnderCommodity
- Group 1
- Group 2
- Group 3
- Group 4
- Group 5
- Group 6
- CCA
- Item Cod
- Commodity Cod Raw
- Commodity Hierarchy
  - Commodity
  - UnderCommodity
  - Group 1
  - Group 2
  - Group 3
  - Group 4
  - Group 5
  - Group 6
  - CCA
  - Item
- Commodity Raw
- Cost Category

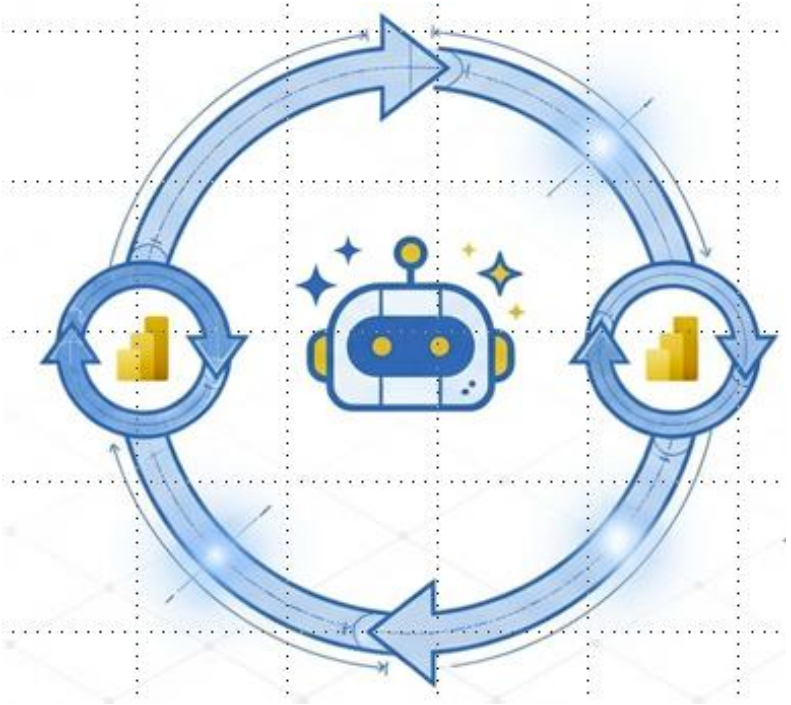
Project-Task | Mktg Focus | Mktg Focus Detail | Report Energie | Project | Last Price | Time Int V2 | Check_Totali | Page 1

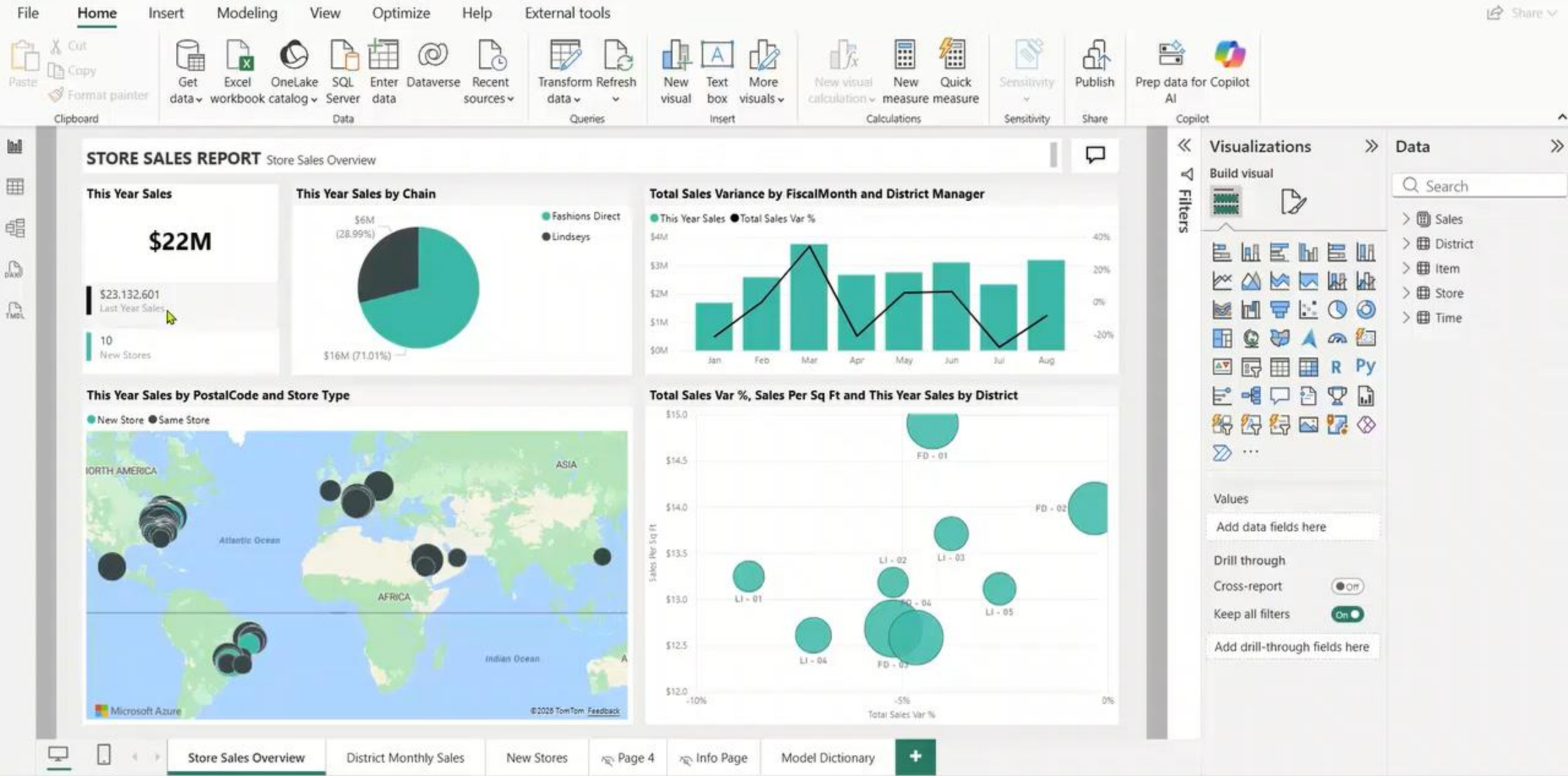
Page 16 of 16

67%

Use Case 2 – Data Dictionary & AI-assisted Automated Documentation






File
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OneLake catalog
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STORE SALES REPORT Store Sales Overview
This Year Sales
$22M
$23.132.601
Last Year Sales
10
New Stores
This Year Sales by Chain
$6M (28.99%)
$16M (71.01%)
Fashions Direct
Lindseys
Total Sales Variance by FiscalMonth and District Manager
This Year Sales
Total Sales Var %
$4M
$3M
$2M
$1M
$0M
40%
20%
0%
-20%
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
This Year Sales by PostalCode and Store Type
New Store
Same Store
NORTH AMERICA
ASIA
Atlantic Ocean
AFRICA
Indian Ocean
Microsoft Azure
©2026 TomTom Feedback
Total Sales Var %, Sales Per Sq Ft and This Year Sales by District
$15.0
$14.5
$14.0
$13.5
$13.0
$12.5
$12.0
Sales Per Sq Ft
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FD - 02
FD - 03
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LI - 05
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Total Sales Var %
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Add drill-through fields here
Data
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Store Sales Overview
District Monthly Sales
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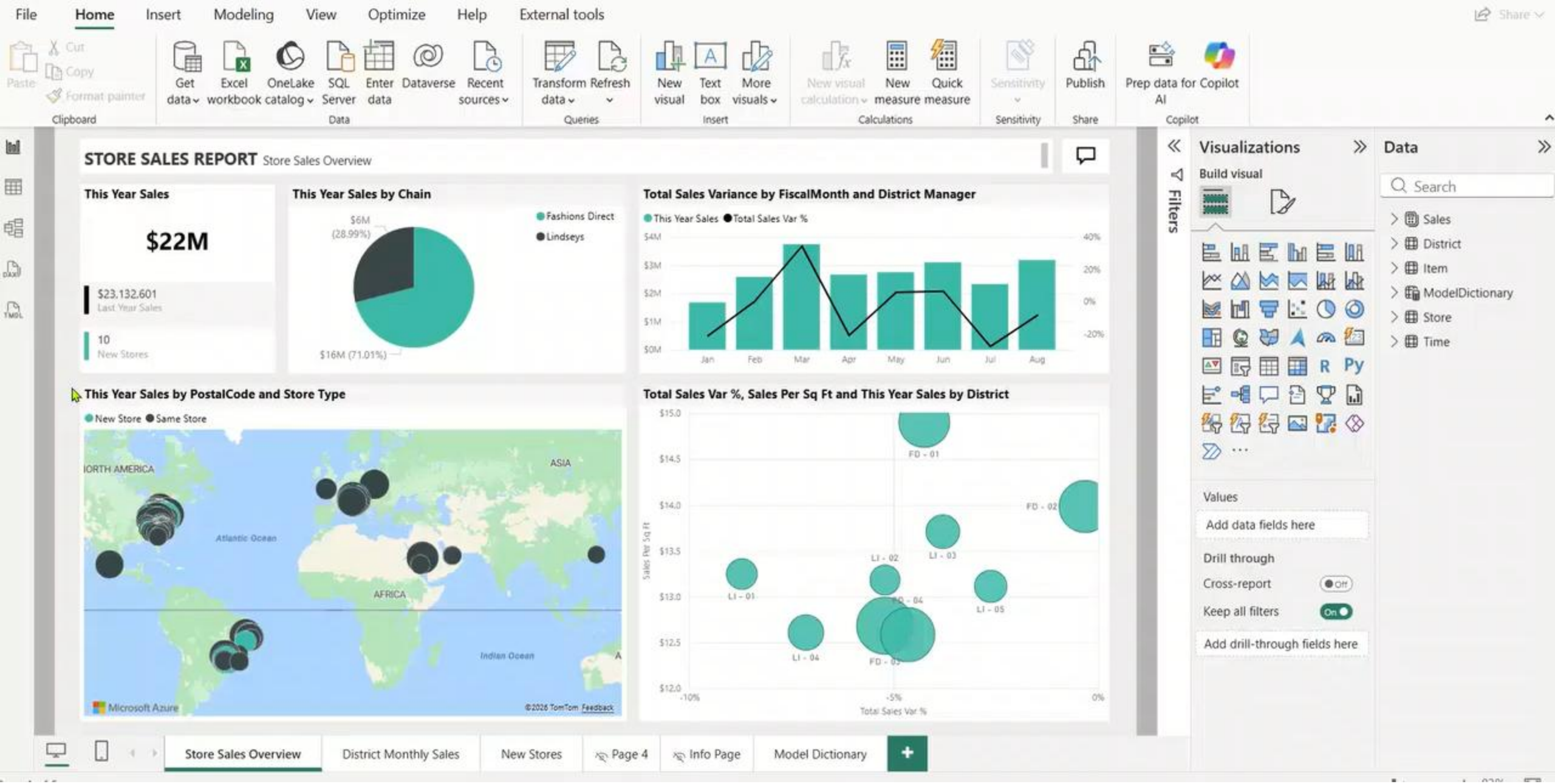
Page 4
Info Page
Model Dictionary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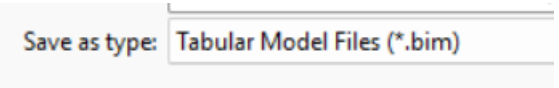
DATA SATURDAYS

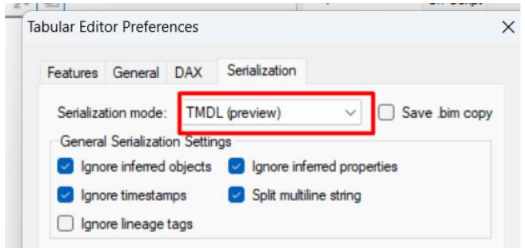
Use Case 3 : Migrate SSAS Legacy to PowerBI SuperFast

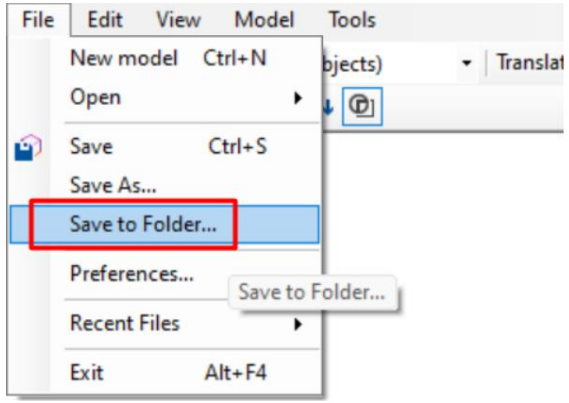
inn0va community
Règolo FARM
CONSORTIVM SERVICE S.r.l.
UNIPORDENONE
moltiplica il valore della formazione

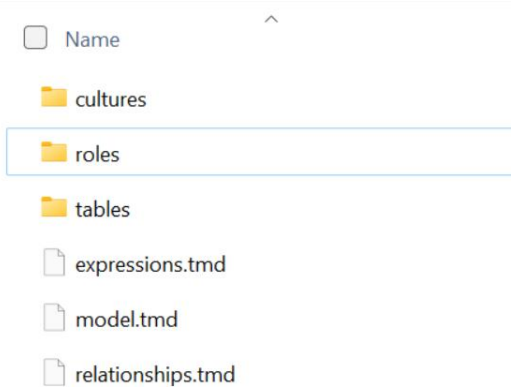
1: Convert SSAS into TMDL using TE

1. Open your SSAS Model in Tabular Editor or Visual Studio and Save file as .bim file.
2. Open Tabular Editor (2 or 3) and set TMDL as serialization mode
3. Reopen .bim file in Tabular Editor -> Save to Folder
4. Now The bim file is serialized into TMDL!



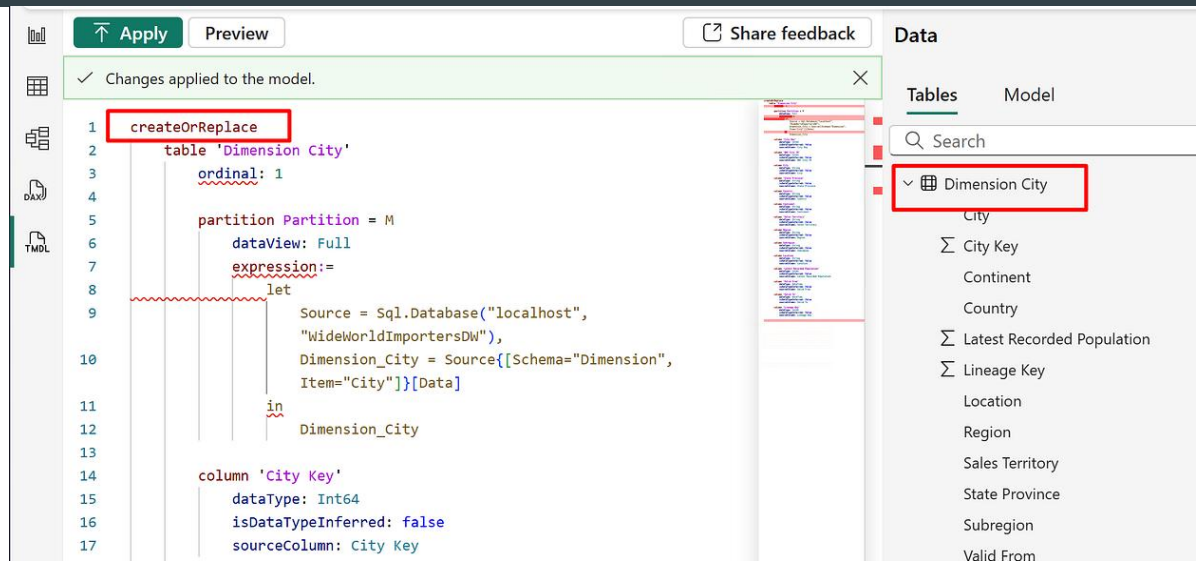


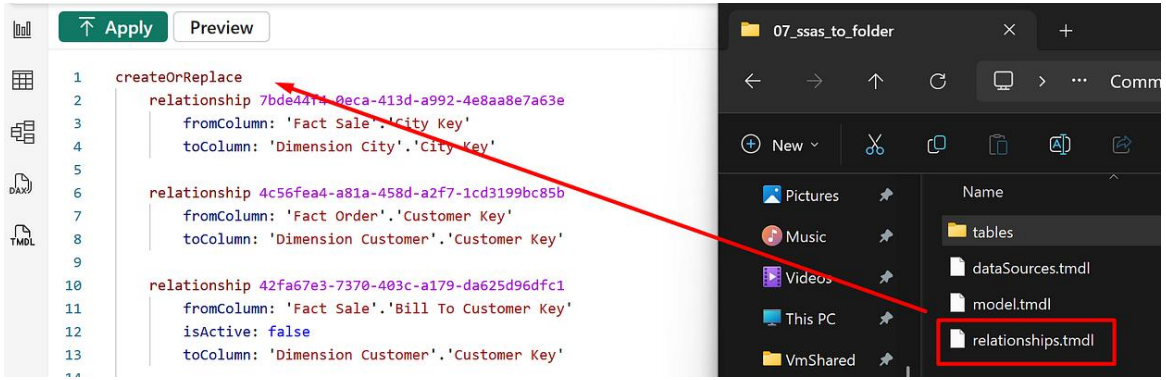




2: Re-Generate Objects in PowerBI using TMDL View

1. Open a New Pbix
2. Create a sample M query as a placeholder and to create the source connection
3. Optional: create Expressions (to supply to query parameters to avoid complete refresh)
4. For each table Copy/paste old TMDL into TMDL view into a CreateorReplace Statement, change partition structure to the new one. Validate code, Apply, Refresh. Repeat for each single table.
5. Script in one shot: relationships, then roles, and remaining objects.
6. Refresh all, publish and…Job done!





DATA SATURDAYS

A new Era for PowerBI Development

PRO

- **Productivity** Massive Edits in seconds
- **Collaboration** Enhance collaboration between teams
- **Transparency** Total control over Metadata
- **Future Proof** Allows AI to edit metadata

CONS

- **Feature Limitations** Lacks some functionality and properties
- **Visuals Risk** Editing via TMDL miss the visuals representation
- **Speed Trap** Massive AI driven editing can corrupt models

What’s your Take about TMDL?